Excel Notes for Analyst Jobs

18 pages. Formulas, pivots, shortcuts.
Every formula tested.

@thedataguy16

Swipe →

1. CELL REFERENCES

THE BASE OF EVERY FORMULA

By @thedataguy16

1. Multiply two cells

```
=A2*B2
```

2. Lock a cell with $

```
=A2*$B$1
```

3. Lock only the row

```
=A2*B$1
```

4. Add a range

```
=SUM(B2:B6)
```

5. Average

```
=AVERAGE(B2:B6)
```

6. Largest value

```
=MAX(B2:B6)
```

7. Count numbers only

```
=COUNT(A2:A6)
```

8. Count filled cells

```
=COUNTA(A2:A6)
```

9. Round to 2 places

```
=ROUND(A2,2)
```

10. Always round up

```
=ROUNDUP(A2,0)
```

2. IF LOGIC

LET EXCEL DECIDE

By @thedataguy16

1 *Pass or fail*

```
=IF(B2>=40,"Pass","Fail")
```

2 *Grade with nested IF*

```
=IF(B2>=75,"A",IF(B2>=60,"B","C"))
```

3 *Same grade with IFS*

```
=IFS(B2>=75,"A",B2>=60,"B",TRUE,"C")
```

4 *Both must be true*

```
=IF(AND(B2>=40,C2>=75),"Eligible","No")
```

5 *Any one true*

```
=IF(OR(D2="Delivered",D2="Closed"),"Done","Open")
```

6 *Is the cell empty?*

```
=IF(ISBLANK(E2),"Missing","Filled")
```

7 *Hide errors*

```
=IFERROR(A2/B2,0)
```

8 *Only hide #N/A*

```
=IFNA(MATCH("Z",A2:A4,0),"Not found")
```

9 *Codes to names*

```
=SWITCH(F2,"N","North","S","South","Other")
```

3. SUM WITH CONDITIONS

SUMIFS RUNS HALF OF MIS

By @thedataguy16

1. *Count one region*

```
=COUNTIF(A2:A7,"North")
```

2. *Count with 2 conditions*

```
=COUNTIFS(A2:A7,"North",C2:C7,"Closed")
```

3. *Sum one region*

```
=SUMIF(A2:A7,"South",B2:B7)
```

4. *Sum with 2 conditions*

```
=SUMIFS(B2:B7,A2:A7,"North",C2:C7,"Closed")
```

5. *Sum values above 250*

```
=SUMIF(B2:B7,">250")
```

6. *Average one region*

```
=AVERAGEIF(A2:A7,"North",B2:B7)
```

7. *Biggest sale in South*

```
=MAXIFS(B2:B7,A2:A7,"South")
```

8. *Smallest open sale*

```
=MINIFS(B2:B7,C2:C7,"Open")
```

9. *Count not closed*

```
=COUNTIF(C2:C7,"<>Closed")
```

10. *Price x qty, summed*

```
=SUMPRODUCT(B2:B7,D2:D7)
```

4. LOOKUPS

VLOOKUP TO XLOOKUP

By @thedataguy16

1. *Exact match*

```
=VLOOKUP(E2,A2:C4,2,FALSE)
```

2. *Third column*

```
=VLOOKUP(E2,A2:C4,3,0)
```

3. *Find a band (slab)*

```
=VLOOKUP(G2,H2:I4,2,TRUE)
```

4. *Position in a list*

```
=MATCH(E2,A2:A4,0)
```

5. *Value at a position*

```
=INDEX(B2:B4,2)
```

6. *INDEX + MATCH*

```
=INDEX(B2:B4,MATCH(E2,A2:A4,0))
```

7. *Look to the left*

```
=INDEX(A2:A4,MATCH("Neha",B2:B4,0))
```

8. *XLOOKUP (Excel 365)*

```
=XLOOKUP(E2,A2:A4,B2:B4)
```

9. *With a not-found value*

```
=XLOOKUP(999,A2:A4,B2:B4,"Not found")
```

10. *Last match, not first*

```
=XLOOKUP(E2,A2:A5,B2:B5,,0,-1)
```

5. CLEAN TEXT

MESSY NAMES, FIXED

By @thedataguy16

1. *Remove extra spaces*

```
=TRIM(A2)
```

2. *Proper case*

```
=PROPER(TRIM(A2))
```

3. *Capital letters*

```
=UPPER(B2)
```

4. *Small letters*

```
=LOWER(B2)
```

5. *First 3 characters*

```
=LEFT(C2,3)
```

6. *Last 4 characters*

```
=RIGHT(C2,4)
```

7. *From the middle*

```
=MID(C2,5,2)
```

8. *Length*

```
=LEN(C2)
```

9. *Position of a dash*

```
=FIND("-",C2)
```

10. *Remove dashes*

```
=SUBSTITUTE(D2,"-","")
```

6. JOIN AND SPLIT

NAMES, EMAILS, LABELS

By @thedataguy16

1. *Join with a space*

```
=A2&" "&B2
```

2. *Same with CONCAT*

```
=CONCAT(A2," ",B2)
```

3. *Join a list, skip blanks*

```
=TEXTJOIN(", ",TRUE,A2:A4)
```

4. *Name before the @*

```
=LEFT(C2,FIND("@",C2)-1)
```

5. *Domain after the @*

```
=MID(C2,FIND("@",C2)+1,50)
```

6. *Text to number*

```
=VALUE(E2)
```

7. *Show as percent*

```
=TEXT(F2,"0.0%")
```

8. *Show a date your way*

```
=TEXT(G2,"dd-mmm-yyyy")
```

9. *Repeat a character*

```
=REPT("*",H2)
```

7. DATES

REPORTS RUN ON DATES

By @thedataguy16

1. Today's date

```
=TODAY()>DATE(2026,1,1)
```

2. Year

```
=YEAR(A2)
```

3. Month number

```
=MONTH(A2)
```

4. Month name

```
=TEXT(A2,"mmm")
```

5. Last day of the month

```
=EOMONTH(A2,0)
```

6. Same day, 3 months later

```
=EDATE(A2,3)
```

7. Months between

```
=DATEDIF(A2,B2,"m")
```

8. Working days between

```
=NETWORKDAYS(A2,B2)
```

9. Weekday (Mon = 1)

```
=WEEKDAY(A2,2)
```

10. 30 days later

```
=A2+30
```

8. EXCEL 365 FORMULAS

NOT IN EXCEL 2016 OR 2019

By @thedataguy16

1. *Each city once*

```
=UNIQUE(A2:A5)
```

2. *Cities seen only once*

```
=UNIQUE(A2:A5,,TRUE)
```

3. *Count distinct cities*

```
=ROWS(UNIQUE(A2:A5))
```

4. *Sort by sales, high first*

```
=SORTBY(A2:C5,C2:C5,-1)
```

5. *Rows with sales over 500*

```
=FILTER(A2:C5,C2:C5>500)
```

6. *Same, with a fallback*

```
=FILTER(A2:C5,C2:C5>5000,"None")
```

7. *Two conditions (AND)*

```
=FILTER(A2:C5,(A2:A5="Pune")*(C2:C5>500))
```

8. *Split at commas*

```
=TEXTSPLIT(D2,",")
```

9. *Split, skip empty bits*

```
=TEXTSPLIT(D2,",",,TRUE)
```

9. PERCENT AND *GROWTH*

EVERY MIS REVIEW ASKS THESE

By @thedataguy16

1. *Share of total*

```
=B2/SUM($B$2:$B$5)
```

2. *Growth vs last month*

```
=(C2-B2)/B2
```

3. *Running total*

```
=SUM($B$2:B4)
```

4. *Rank, highest = 1*

```
=RANK.EQ(B3,$B$2:$B$5)
```

5. *Second highest*

```
=LARGE(B2:B5,2)
```

6. *Lowest*

```
=SMALL(B2:B5,1)
```

7. *Yearly growth over 3 years*

```
=(C2/B2)^(1/3)-1
```

8. *Percent as a label*

```
=ROUND(B2/C2*100,1)&"%"
```

9. *Growth, safe from zero*

```
=IFERROR((C2-B2)/B2,"")
```

10. SPOT DIRTY DATA

CHECK BEFORE YOU ANALYSE

By @thedataguy16

1. *Flag duplicates*

```
=COUNTIF($A$2:$A$6,A2)>1
```

2. *Label duplicates*

```
=IF(COUNTIF($A$2:$A$6,A2)>1,"Duplicate","")
```

3. *Is it blank?*

```
=ISBLANK(B2)
```

4. *Is it a real number?*

```
=ISNUMBER(C2)
```

5. *Fix number-as-text*

```
=VALUE(TRIM(C2))
```

6. *Remove line breaks*

```
=CLEAN(D2)
```

7. *Count blank cells*

```
=COUNTBLANK(B2:B6)
```

8. *Extra spaces hiding*

```
=LEN(F2)-LEN(TRIM(F2))
```

9. *Same text, same case?*

```
=EXACT(G2,H2)
```

11. Pivot tables, step by step

by @thedataguy16

#	Step	How
1	Turn data into a table	`Click inside the data, Insert > Table`
2	Insert the pivot	`Insert > PivotTable > New Worksheet`
3	Sales by region	`Drag Region to Rows, Sales to Values`
4	Split by month	`Drag Month to Columns`
5	Count instead of sum	`Value Field Settings > Count`
6	Show % of total	`Show Values As > % of Grand Total`
7	Group dates by month	`Right-click a date > Group > Months`
8	Filter one region	`Drag Region to Filters`
9	Clickable filter buttons	`PivotTable Analyze > Insert Slicer`
10	Biggest first	`Right-click a value > Sort > Largest to Smallest`
11	Show only the best 5	`Row label filter > Value Filters, set it to 5`
12	New data added	`Right-click the pivot > Refresh`
13	Blanks should show 0	`PivotTable Options > For empty cells show: 0`
14	Make a chart from it	`PivotTable Analyze > PivotChart`
15	See rows behind a number	`Double-click that number`
16	Flat, report-style layout	`Design > Report Layout > Show in Tabular Form`

12. CONDITIONAL FORMATTING

RULES THAT USE A FORMULA

By @thedataguy16

1. *Sales above 1000*

```
=$C2>1000
```

2. *Pending orders*

```
=$D2="Pending"
```

3. *Duplicates in column A*

```
=COUNTIF($A:$A,$A2)>1
```

4. *Highest value*

```
=C2=MAX($C$2:$C$10)
```

5. *Every other row*

```
=MOD(ROW(),2)=0
```

6. *Overdue dates*

```
=$E2<TODAY()
```

7. *Empty cells*

```
=ISBLANK($B2)
```

8. *Between 500 and 999*

```
=AND($C2>=500,$C2<1000)
```

9. *Weekend dates*

```
=WEEKDAY($E2,2)>5
```

10. *Text contains delhi*

```
=ISNUMBER(SEARCH("delhi",$F2))
```

13. MIS REPORT FORMULAS

MONTHLY REVIEW, SORTED

By @thedataguy16

1. *Month total*

```
=SUMIFS(C:C,B:B,"Jan")
```

2. *Region + month*

```
=SUMIFS(C:C,A:A,"North",B:B,"Jan")
```

3. *Orders in a month*

```
=COUNTIFS(B:B,"Mar")
```

4. *Share of one region*

```
=SUMIF(A:A,"North",C:C)/SUM(C:C)
```

5. *Target met?*

```
=IF(D2>=E2,"Met","Missed")
```

6. *Achievement %*

```
=D2/E2
```

7. *Month on month*

```
=D3/D2-1
```

8. *Weighted average price*

```
=SUMPRODUCT(F2:F4,G2:G4)/SUM(G2:G4)
```

9. *Year to date*

```
=SUM($D$2:D4)
```

10. *Gap to target*

```
=MAX(0,E2-D2)
```

14. Which chart, when

by @thedataguy16

#	You want to show	Use
1	Change over months	`Line chart`
2	Compare a few categories	`Column chart`
3	Long category names	`Horizontal bar chart`
4	Rank many items	`Bar chart, sorted`
5	Share of a whole, 2 to 4 parts	`Pie or doughnut`
6	Share, month by month	`Stacked column`
7	Two measures, different scales	`Combo chart with a secondary axis`
8	Link between two numbers	`Scatter chart`
9	How values are spread	`Histogram`
10	Ups and downs to a final total	`Waterfall chart`
11	One headline number	`A big number with a label`
12	Actual vs target	`Column with a target line`
13	Few causes, most of the effect	`Pareto chart`
14	Sales by state	`Map chart (Excel 365)`

15. Shortcuts that save time

by @thedataguy16

#	Do this	Keys (Windows)
1	Make a table	`Ctrl + T`
2	Filters on or off	`Ctrl + Shift + L`
3	AutoSum	`Alt + =`
4	Flash Fill	`Ctrl + E`
5	Quick chart	`Alt + F1`
6	Chart on a new sheet	`F11`
7	Format Cells	`Ctrl + 1`
8	Add $ to a reference	`F4`
9	Fill down	`Ctrl + D`
10	Edit the cell	`F2`
11	New line inside a cell	`Alt + Enter`
12	Select the column	`Ctrl + Space`
13	Select the row	`Shift + Space`
14	Insert today's date	`Ctrl + ;`
15	Paste Special	`Ctrl + Alt + V`
16	Jump to the edge of data	`Ctrl + Arrow key`

16. Errors and how to fix them

by @thedataguy16

#	Error	What it usually means
1	#N/A	Lookup value not found. Check spaces, text vs number.
2	#VALUE!	Text where a number was expected.
3	#REF!	The formula points to a deleted cell.
4	#DIV/0!	Dividing by zero or a blank cell.
5	#NAME?	Misspelt function, missing quotes, or not in your version.
6	#SPILL!	A 365 formula has no room. Clear the cells below.
7	#NUM!	An impossible number, like a bad date sum.
8	#####	Column too narrow. Widen it.
9	Total looks wrong	Numbers stored as text. They sit on the left.
10	VLOOKUP picks a wrong row	Last argument missing. Use FALSE.
11	Formula not updating	Formulas > Calculation Options > Automatic.
12	Circular reference	The formula uses its own cell.

17. Interview questions, short answers

by @thedataguy16

#	Question	Answer
1	VLOOKUP vs XLOOKUP?	XLOOKUP can look left, has a not-found value and is exact by default.
2	What does $ do in $A$1?	Locks the column and row when you copy.
3	COUNT vs COUNTA?	COUNT: numbers only. COUNTA: any filled cell.
4	Remove duplicates?	Data > Remove Duplicates, or UNIQUE in 365.
5	SUMIF vs SUMIFS?	SUMIF: one condition. SUMIFS: many, sum range first.
6	What is a pivot table?	A drag-and-drop summary. No formulas needed.
7	Find duplicates, keep data?	COUNTIF(range, cell)>1 or highlight rules.
8	What is Flash Fill?	Ctrl + E. Excel repeats the pattern you typed.
9	Why INDEX + MATCH?	Looks both ways and survives inserted columns.
10	How do you handle errors?	Fix the cause, then IFERROR or IFNA.
11	Relative vs absolute?	Relative moves when copied. Absolute stays.
12	What is data validation?	Rules for what a cell accepts, like a drop-down.
13	Freeze the header row?	View > Freeze Panes > Freeze Top Row.
14	What is Power Query for?	Repeatable cleaning: import, clean, combine, refresh.

18. Practise these 12 tasks

by @thedataguy16

#	Task	You will use
1	Total sales by region and month	`SUMIFS or a pivot`
2	Fill a customer's city from another sheet	`XLOOKUP or INDEX + MATCH`
3	Clean 500 messy names	`TRIM, PROPER`
4	Split emails into name and domain	`LEFT, MID, FIND`
5	Flag duplicate invoices	`COUNTIF`
6	Grade students into A/B/C	`IFS`
7	Month-on-month growth table	`Percent formulas`
8	Overdue payments in red	`Conditional formatting`
9	Best 5 products by revenue	`SORTBY or a sorted pivot`
10	Working days per order	`NETWORKDAYS`
11	One-page sales dashboard	`Pivot + slicer + chart`
12	Target vs actual report	`IF, percent, column chart`

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